Research of New Super Transformer Architecture – the Proposal of Master's Thesis and Pilot Study

Mateusz Szkopek

490228

matszk3@st.amu.edu.pl

The year 1950 and the publication of Turing's "Computing Machinery and Intelligence" can be regarded as the beginning of the era of artificial intelligence (AI). When analyzing this seminal text, most philosophers focus primarily on the famous Turing Test, intended to demonstrate a machine's capacity for thought, as well as on Turing's extensive argumentation for the possibility that machines might possess such a capacity. Much less attention, however, is paid to his description of what such a machine and the process of its learning might actually look like. Turing points out that it may be difficult to create a program that immediately imitates the mind of an adult human and that it would be better to begin by creating a child's mind, together with rules governing its education (Turing, 1950). This idea is widespread in machine learning (ML), including deep learning (DL) and the currently dominant transformer architecture (Vaswani et al., 2017), one variant of which is the decoder-only architecture characterized by a masked self-attention mechanism that restricts each token's access to preceding tokens in the sequence and is frequently used in autoregressive models, such as models of the generative pretrained transformer (GPT) class. The learning process as envisioned by Turing, however, looks entirely different. According to the original conception, the child machine would learn continuously (instant learning) through incessant interaction with a human teacher, which could lead to the machine acquiring numerous individual experiences and perhaps even developing an emotional teacher–student or parent–child relationship with that person. Moreover, every sentence or instruction uttered by the teacher was supposed to be immediately incorporated by the machine, without, for example, 512 epochs of repetition, into a body of well-established truths, implying a form of immediate and deterministic learning (Turing, 1950). In such a process, what could be particularly important is that, just like human children, the machine would have to continually interpret the intentions underlying the teacher's instructions. According to currently dominant theories, this is crucial both for acquiring language with genuine understanding and for developing a theory of mind. Overall, this older model of machine learning evokes numerous associations with modern approaches to education that emphasize understanding, creativity, relationships, and individualized attention to the learner, and we suspect that it might ultimately even lead to the emergence of artificial general intelligence (AGI), understood as artificial intelligence that matches or surpasses the average human in all types of intellectual work. This is because, in such work, a deep understanding of the intentions behind a superior's instructions is far more important than literally interpreting and carrying out every word they say, while new and previously unforeseen outcomes—for example, in scientific research—may require a highly individualized, creative, and novel approach.

By contrast, currently dominant transformer architectures in DL, first described in the seminal paper "Attention Is All You Need" by Vaswani et al. in 2017, based on the attention mechanism and still used by most recent studies, learn only during the training phase, after which their weights are frozen. This learning occurs primarily through repeated exposure to the same texts over many training epochs, with the information being progressively encoded in the model's weights, which may account for their lack of genuine language understanding and for the fact that the loss landscapes of their neural networks exhibit features characteristic of the brains of individuals with Wernicke's aphasia (Watanabe et al., 2025). Even in reinforcement learning from human feedback (RLHF) techniques, the machine is typically trained by many, often anonymous individuals, who reward it not for interpreting their intentions or for deeply understanding the subject matter, but simply for producing an answer they find satisfactory (which is not necessarily correct). This is likely a major cause of problematic phenomena such as sycophancy and hallucinations. Moreover, when attempts are made during fine-tuning to teach an already trained model new things, as in Savigny & Yun (2026), this is usually achieved by manually adding further layers to the model and applying a conventional supervised-learning process involving the network predicting an answer, calculating the loss, and performing backpropagation. Overall, such contemporary training methods evoke associations with the old Prussian educational system, implemented under Bismarck, which was based on repeated drilling without deeper understanding until all the required information had been memorized.

This raises the question of whether we can replace the currently dominant "Prussian educational system" of models with the much more modern system proposed by Turing in 1950. It seems that the aforementioned *instant learning* may be particularly important for implementing such an approach, as it is what makes it possible to retain individual interactions between a human and a machine. However, a major problem arises when attempting to implement it: during *instant learning*, standard static transformer models (Vaswani et al., 2017) either immediately forget previously acquired knowledge by overwriting their weights, when historical training data are not replayed, or, when they are instructed to combine historical data with new data and repeatedly train on the entire dataset, eventually prove to have too few parameters, resulting in a decline in performance. Since the transformer architecture appears to lack the potential for learning according to Turing's proposed system—and thus for achieving AGI—we consider it necessary to search for new neural architectures that are better suited to this purpose. Full *instant learning* at the human level should, in turn, lead to AGI, since in the definition given above we have defined this level as the standard or higher cognitive abilities of an adult human. Every child reaches this level, so a learning system that learns in the same way as human children should, in principle, be capable of reaching it as well. Many researchers share this view, although they have generally not attempted to connect it with the architectural examples discussed below. We absolutely do not claim that we will achieve this level in our work. Rather, our aim is to establish a new foundation on which such systems can subsequently be built and scaled.

One fact that significantly facilitates this goal is that the human brain already learns in this way, so appropriately replicating some of its mechanisms should produce promising results. The scientific community has, in fact, already begun research into such biomimetic architectures. One of the most promising approaches appears to be *infomorphic neurons*, inspired by the information-processing structure of pyramidal cells in the human brain, initially proposed in a version with two types of inputs (Makkeh et al., 2025) and subsequently extended to three (Schneider et al., 2025). A particular advantage of this approach is its high degree of interpretability, which could provide a starting point for deep representation engineering and for the development of more deterministic architectures. The authors of the concept, however, have not yet tested it either in transformers or in instant-learning models, which we intend to do in our research.

Although infomorphic neurons may make an important contribution to reducing sycophancy and hallucinations, their static architecture may not be sufficient for fully continuous learning. A promising way of complementing them, however, may be the concept of neurogenesis, likewise derived from neuroscience. Numerous example implementations in multi-layer perceptron (MLP) and convolutional neural network (CNN) architectures were presented by Maile et al. (2022), with the NORTH-Select method appearing particularly promising. We therefore also intend to employ this method.

The use of neurogenesis alone may nevertheless still result in the development of networks with performance comparable to that of current architectures. Compared with

MLPs/transformers, the brain achieves substantially greater efficiency despite its neurons not being fully interconnected. One of the most important processes responsible for this organization of its connectome appears to be the mechanism of adaptive rewiring, which is discussed extensively in Li et al. (2024).

Other lines of research that may contribute to overcoming the limitations of transformers include, among others, research on combining them with capsule layers, represented by Yu et al. (2024), and research on the ways in which representations are formed in neural networks, represented by Ziyin et al. (2025). Our primary objective in the present study will be to test the first two approaches (or three, if time permits) in combination with the transformer architecture and thereby attempt to develop a super transformer, defined here as an architecture modeled on the transformer but equipped with a greater number of mechanisms and possessing capabilities beyond those of currently used architectures. Due to time and resource constraints, the remaining approaches will therefore be explored to a lesser extent—or not at all. We of course leave open the possibility of returning to them at a later stage or as part of a separate study.

Since this is architectural research on DL, it is necessary to precisely define the task on which the architectures under investigation will be compared, as well as the criteria for success. Unfortunately, due to the computational costs and financial constraints involved, we are not yet able to develop a generative language model capable of competing with frontier models such as GPT-5.6 Sol or the as-yet-unreleased Astra across a broad range of popular benchmarks, nor of independently making mathematical breakthroughs like the latter. Instead, we chose a classification task that is commonly encountered in ML. The primary objective of the study is to determine whether the proposed experimental model architectures can achieve higher performance on the test set than models with standard architectures and the same or greater model size. Model performance will be evaluated using three classification metrics: Accuracy, Macro-F1, and Weighted-F1. In addition to predictive performance, training time will be considered as a secondary criterion, with shorter training times being preferred when the models achieve comparable or better results. More specifically, the task will fall within the field of natural language processing (NLP) and will involve text classification. It will be similar in nature to the task used by Yu et al. (2024) to compare different model architectures. However, in our study, the classification task will specifically concern argument mining (AM). The proposed architectures will therefore be considered successful if they outperform the corresponding standard-architecture models in terms of the selected evaluation metrics (mainly Macro-F1) while maintaining a comparable or lower training time.

Argument mining (AM) is a relatively young but rapidly developing field of study concerned with how to identify arguments in text, segment them into components and classify those components, and reconstruct entire argumentative structures in as automated a manner as possible while maintaining the highest possible quality. Its methods draw on a variety of traditions whose evolution closely reflects the broader history of NLP, beginning with classical methods based on searching texts for specific words, expressions, or linguistic constructions, followed by classical ML techniques such as k-nearest neighbors (kNN), support vector machines (SVMs), and decision trees, then classical DL approaches involving MLPs, CNNs, and recurrent neural networks (RNNs), and finally transformers and large language models (LLMs). Today, the main challenges in AM include, above all, the lack of a single universally accepted theory of argumentation, the quality of annotations in datasets, hallucinations, and the low interpretability of the most advanced models. The latter obstacle could, however, be partially mitigated through the use of the aforementioned infomorphic neurons (Makkeh et al., 2025; Schneider et al., 2025).

In recent years, numerous studies have addressed various aspects of AM. For example, Gemechu et al. (2025) developed an open-source, scalable, and modular framework integrating numerous methods commonly used in AM pipelines. Particularly importantly, given the lack of consensus in the field, the framework operates independently of the methodology and theory of argumentation adopted. The previously mentioned Savigny & Yun (2026), in response to the high degree of fragmentation among AM data, developed a unified dataset comprising 19 previously existing datasets. They then identified eight tasks that can be defined on this unified dataset and investigated whether a single fine-tuned LLM could be used effectively for all eight tasks. Interestingly, the model fine-tuned jointly on all eight tasks achieved even higher performance than models fine-tuned for individual tasks, although its training cost was relatively high. A potential compromise between cost and performance was to train the models separately and subsequently combine them through model merging. The high performance of the model fine-tuned on all tasks may be interpreted as evidence that different AM tasks are not entirely independent and partially overlap. Bai et al. (2026), in turn, argue that different types of texts may require different theoretical approaches. They base their findings on an analysis of their own corpus of German-language essays written by ninth-grade secondary-school students (and therefore often less professionally written texts), which demonstrated that the standard representation of argumentative structure as a tree was inadequate for these essays. In addition, they conducted a study on their corpus examining the quality of automatic classification of argumentative components by LLMs depending on the prompt, demonstrating that models can support the annotation process and that a few-shot approach outperformed alternatives even with a very small number of examples. Uberna et al. (2026), meanwhile, analyzed the capabilities of zero-shot agents and theoretically grounded agents using a retrieval-augmented generation (RAG)-based approach for the automatic classification of the pragmatic and rhetorical functions of reformulations. The latter achieved a Macro-F1 score nearly 30% higher than that of the zero-shot agents, indicating the particular importance of theoretical knowledge in such tasks. A better understanding and representation of this knowledge could likely lead to even higher performance. Finally, it is worth mentioning the work of Pietroń et al. (2026), who compared various LLMs on an argument-classification task. GPT-5.2, GPT-oss-120B, and GPT-oss-20B ranked first, second, and third, respectively, in most variants, outperforming DeepSeek and Llama models. Moreover, the study examined model performance under different prompting strategies and demonstrated the superiority of systems composed of multiple model instances that voted on the class to which a given argument belonged over models prompted using zero-shot, manual rephrase and respond (mRAR), and chain-of-thought (CoT) techniques. The authors also conducted qualitative error analyses, which surprisingly revealed that not only the models but also a substantial proportion of the original human labels were incorrect. This provides particularly strong evidence of the importance of precise annotation: a model trained on incorrectly labeled or low-quality data may never learn to perform the task correctly, regardless of its architecture. Our study will thus be situated within this broader context of intensive research on AM, although it will approach the topic primarily from the perspective of model architecture.

As part of the preparations for this study, we conducted a pilot experiment using classifiers based on encoder-only transformers, although in the main study we would also like to train and evaluate decoder-only generative models, as this has been a popular direction of research in recent years and, equally importantly, constitutes the main area of research for further study after the completion of a master's degree. The experiment was conducted on a dataset of 6,203 instances containing various types of arguments extracted from comments on Reddit posts and YouTube videos concerning American patriotism, all in English. The data were collected manually and subsequently annotated by seven human annotators as part of the PersOn project, led by Dr. Barbara Konat at the Laboratory of Everyday Argumentation and Persuasion (LEAP) laboratory at the Faculty of Psychology and Cognitive Sciences of Adam Mickiewicz University in Poznań. All training data, together with the complete dataset after initial preprocessing and the code used for its extraction, are available on the lead author's GitHub repository, as is the code used to train the models, along with detailed results and the values of configuration variables used in the experiments, at https://github.com/MateuszSzkopek/Research-of-New-Super-Transformer-Architecture. To properly characterize these materials, it is worth noting above all that they were divided into training and test subsets in an 80:20 ratio, while the full dataset contained 4,549 examples of Asserting, 599 examples of Analysing, 386 examples of Emotion-appealing, 275 examples of Emotion-expressing, 151 examples of Rhetorical Questioning, 84 examples of Pure Questioning, 81 examples of Phatic-maintaining, and only 78 examples of Assertive Questioning in the text and annotation columns. These figures demonstrate how unevenly the argumentative components of different types were distributed and how few instances were available overall. Although this may adversely affect the ultimate classification performance of the models, paradoxically, it makes the dataset an excellent testbed for different encoder architectures: the task is difficult, meaning that only a genuinely strong architecture is likely to achieve a high score on it. Of course, in the main study, a separate validation subset will be used to select the best-performing model (using an 80:10:10 split). Here, we did not set aside a separate validation subset solely because of the pilot nature of the study and the very small number of examples, particularly for some classes. We adopted a deliberately conservative approach to preprocessing and decided against removing stopwords, lemmatizing, or stemming the texts. Our goal was for the models to learn language as naturally as possible while also having access to as much information as possible from which they could learn.

The study consisted of three experiments (0–2). The aim of Experiment 0 was to empirically determine suitable hyperparameters for the Encoder_BASELINE transformer, which had previously been implemented in TensorFlow. We did not aim to identify the best possible hyperparameter configuration overall. Instead, we sought only a sufficiently good configuration that would provide a reasonably ambitious baseline. Of the 18 models (0–17) that were trained, we selected the one with the highest Macro-F1 score on the test subset. The winner was Model 0, with a score of approximately 0.3073, an ff_dim of 128, a d_model of 16, one attention head, and a single block. The fact that such a small model emerged as the winner may be due to the relatively small amount of training data available for larger architectures to fit to. It is also worth noting the extremely short training time, which is highly beneficial when conducting numerous experiments.

Experiment 1 involved modifying Model 0 from Experiment 0 by replacing its Dense neurons with InfomorphicDense neurons, based on the work of Makkeh et al. (2025) and Schneider et al. (2025). To this end, we developed our own TensorFlow-based implementation (with the assistance of OpenAI Codex), using the same framework employed throughout our research. The additional trivariate=False argument allows the neurons to use a third type of input, namely lateral inputs (in addition to receptive and contextual inputs), when set to trivariate=True, although in the pilot study we used exclusively the two-input variant of the neuron. Merely replacing the first layer of the feed forward network (FFN) with InfomorphicDense(ff_dim, activation="relu", trivariate=False) increased the Macro-F1 score of Model 0 in Experiment 1 to approximately 0.3131. Unfortunately, the next two models did not yield any further improvement, which may suggest that infomorphic neurons perform best at earlier stages of processing. Of course, the very small training dataset, together with the equally small model, limits the scope of this conclusion; nevertheless, this issue warrants further investigation in future research.

Experiment 2 was conducted in a very similar manner, although in designing the NeurogenesisDense layer, we based our approach on the work of Maile et al. (2022). With the assistance of Codex, we developed a TensorFlow-based neuron class utilizing the neurogenesis mechanism according to the NORTH-Select method, which was reported by the authors to be the most effective of the methods they investigated. Unfortunately, in our study, none of the six models outperformed Model 0 from either of the two previous experiments. Contrary to the findings of Maile et al., neurogenesis tended to reduce rather than improve model performance. This negative result encourages further investigation in the future, as the mechanism itself still appears to hold considerable potential.

Risk	Probability	Severity	Contingency measure
The proposed architectural modifications do not outperform the standard transformer baseline.	Medium	High	Treat the absence of improvement as a valid experimental result and conduct a detailed ablation and error analysis to determine which components limit performance. The study will compare different placements and combinations of the proposed mechanisms rather than relying on a single configuration. If no improvement is obtained, the final thesis will focus on identifying the conditions under which the modifications fail and on comparing their computational and representational properties with the baseline.
The available social-media dataset is insufficient for reliable training or evaluation.	High	Medium	Use the existing PersOn dataset as the initial experimental benchmark and monitor the effect of dataset size and class imbalance on model performance. The dataset currently contains 6,203 instances, with substantial imbalance and fewer than 100 examples for some classes. If additional data are required, the study will extending or combining suitable argument-mining datasets.
Strong class imbalance prevents meaningful evaluation of the proposed architectures.	High	Medium–High	Use Macro-F1 as the main performance indicator, complemented by Accuracy and Weighted-F1, and report class-level performance and error analysis. A separate validation set will be used in the main study to reduce the risk of selecting models based on the test set.

The planned architectural modifications are too computationally expensive for the available resources.	Medium	High	Prioritize the two main approaches already identified in the proposal—infomorphic neurons and neurogenesis—and treat adaptive rewiring and decoder-only models as optional extensions. This preserves a complete core research project even if the more computationally demanding experiments cannot be conducted.
The broader AGI-related assumptions cannot be operationalized within a Master's thesis.	Very high	Low	Keep AGI as the long-term motivation rather than the direct success criterion of the study. The empirical objective will remain narrowly defined as comparing the proposed architectures with standard architectures on an NLP argument-mining classification task using Macro-F1, Accuracy, Weighted-F1, and training time.
The proposed architecture improves predictive performance but requires substantially more training time.	Medium	Medium	Treat training time as a secondary evaluation criterion and investigate whether the performance gain justifies the additional computational cost. Configurations with comparable predictive performance and lower training time will be preferred.

Table 1: Risk analysis and contingency measures

In conclusion, AI as a scientific discipline appears to have encountered, with the emergence of transformers, a barrier preventing its further effective development and the creation of models that genuinely learn in the way human children do (Turing, 1950). Contemporary research is attempting to overcome this obstacle by developing increasingly advanced architectures, and our work forms part of this broader trend. Progress in AI also directly translates into progress in AM, whose challenges we seek to address here. In the practical part of the study, we trained and evaluated a total of 27 different models, including three infomorphic transformers and six transformers incorporating neurogenesis. While we were able to confirm the positive impact of the former technique, the same could not be said of the latter. Ultimately, we achieved an improvement of only about 0.58 percentage points over the original result, but we view this only as motivation to make even greater efforts. The analysis of the risks associated with the project and the preventive measures have been presented in Table 1 above. We intend to continue our research.

Chapter Outline and Resulting Plan for Further Work:

1. Introduction – A broader search of the scientific literature should be conducted, together with a more thorough examination of the theory underlying the design of the most advanced systems. It would also be worthwhile to review a larger number of texts on AM in order to gain a better understanding of the field in which the models will be tested. I plan to devote the first semester to this work.

2. Methods – During the second semester, all experiments worth conducting should be planned and described in great detail. To this end, I will familiarize myself more closely with best methodological practices and base the experimental design on the literature I have reviewed. In general, I intend to continue working on encoder-only infomorphic transformers and the neurogenesis mechanism in such models, as well as to combine the two approaches. If time permits, I may also implement a third technique—adaptive rewiring. In addition, I would like to train a decoder-only (GPT-style) model as soon as I have sufficient computational resources and an adequate dataset for such an experiment to be meaningful.

3. Results – This chapter will be developed after the planned research has been conducted during the third semester. In addition to carrying out the experiments themselves, I will simultaneously perform preliminary processing and analysis of the results obtained.

4. Discussion – To prepare this chapter, during the fourth semester I intend to discuss the research findings in greater detail and draw conclusions from them, which I will compare with the findings of the most recent literature. In parallel, I will be able to optimize the models' performance as part of my preparations for doctoral studies.

5. Conclusion – I will write this chapter during the summer following the final semester. Since it will be relatively short, I will have plenty of time to prepare for the thesis defense and subsequently for doctoral studies.

6. Bibliography – This section will be compiled throughout the work described above, alongside the collection and systematic organization of the relevant literature. For this reason, the first semester will be a particularly important period for gathering sources.

Bibliography:

1. Turing, A. M. (1950). Computing machinery and intelligence. Mind, 59(236), 433–460.
2. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, & R. Garnett (Eds.), Advances in Neural Information Processing Systems (Vol. 30, pp. 5998–6008). Curran Associates.
3. Makkeh, A., Graetz, M., Schneider, A. C., Ehrlich, D. A., Priesemann, V., & Wibral, M. (2025). A general framework for interpretable neural learning based on local information-theoretic goal functions. Proceedings of the National Academy of Sciences, 122(10), e2408125122. https://doi.org/10.1073/pnas.2408125122
4. Schneider, A. C., Neuhaus, V., Ehrlich, D. A., Makkeh, A., Ecker, A. S., Priesemann, V., & Wibral, M. (2025). What should a neuron aim for? Designing local objective functions based on information theory. ICLR 2025. https://openreview.net/forum?id=CLE09ESvul
5. Watanabe, T., Inoue, K., Kuniyoshi, Y., Nakajima, K., & Aihara, K. (2025). Comparison of large language model with aphasia. Advanced Science, 12(22), e2414016. https://doi.org/10.1002/advs.202414016
6. Maile, K., Rachelson, E., Luga, H., & Wilson, D. G. (2022). When, where, and how to add new neurons to ANNs (arXiv:2202.08539). arXiv. https://arxiv.org/abs/2202.08539
7. Li, J., Bauer, R., Rentzeperis, I., & van Leeuwen, C. (2024). Adaptive rewiring: A general principle for neural network development. Frontiers in Network Physiology, 4, Article 1410092. https://doi.org/10.3389/fnetp.2024.1410092
8. Yu, X., Luo, S.-N., Wu, Y., Cai, Z., Kuan, T.-W., & Tseng, S.-P. (2024). Research on a capsule network text classification method with a self-attention mechanism. Symmetry, 16(5), 517. https://doi.org/10.3390/sym16050517
9. Ziyin, L., Chuang, I. L., Galanti, T., & Poggio, T. (2025). Formation of representations in neural networks. Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025). https://proceedings.iclr.cc/paper_files/paper/2025/hash/ab6a2c6ee757afe43882121281f6065c-Abstract-Conference.html
10. Pietroń, M., Gampel, F., Gomułka, J., Tomski, A., & Olszowski, R. (2026). A comprehensive study of LLM-based argument classification: from Llama through DeepSeek to GPT-5.2 (arXiv:2603.19253). arXiv. https://doi.org/10.48550/arXiv.2603.19253
11. Gemechu, D., Ruiz-Dolz, R., Górska, K., Moslemnejad, S., Maguire, E., Zografistou, D., Jo, Y., Lawrence, J., & Reed, C. (2025). The open argument mining framework. In P. Mishra, S. Muresan, & T. Yu (Eds.), Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations) (pp. 318–328). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.acl-demo.31
12. Savigny, H., & Yun, B. (2026). AMELIA: A family of multi-task end-to-end language models for argumentation. Argument & Computation, 17(1), 79–103. https://doi.org/10.1177/19462174251413169
13. Bai, X., Afzal, K., Benndorf, D., Deutzmann, L., Giera, W.-K., Graßnick, E., & Stede, M. (2026). From newspapers to classrooms: Adapting an annotation scheme and automatic classifiers to mixed-quality argumentative school essays. Argument &

Computation. Advance online publication. https://doi.org/10.1177/19462174261466578
14. Uberna, M., Wawer, M., Chudziak, J. A., & Koszowy, M. (2026). On theoretically-driven LLM agents for multi-dimensional discourse analysis. In A. P. Rocha, L. Steels, & J. van den Herik (Eds.), Proceedings of the 18th International Conference on Agents and Artificial Intelligence (ICAART 2026) (Vol. 1, pp. 385–392). SciTePress. https://doi.org/10.5220/0014319700004052